Beginnings of Classical Physics

1400 – 1600

Natural Sciences

Before

There were problems with GeoCentric Model as Math did not agree with Observations. However, when clash occurred the religious scripture prevailed.

Greek, Indian, and Muslim scholars had proposed a heliocentric (sun centered) solar system, but Western scholars continued the Aristotelian philosophy that the Earth was the center of the universe.

Why hard ?
Relativity - Ordinary

Example: On a moving ship, a cannon ball is dropped from a mast.

If observed from the shore, the cannon ball will have two motions – a forward motion provided by the ship, and a downward motion caused by gravity.

If observed from the ship, the cannon ball will have only one motion – the downward motion caused by gravity.

What is Galilean Relativity?

Copernicus – 1473-1543

Nicholaus Copernicus (born in Poland) was educated as a mathematician and lived during the Reformation when the Christian church became divided.

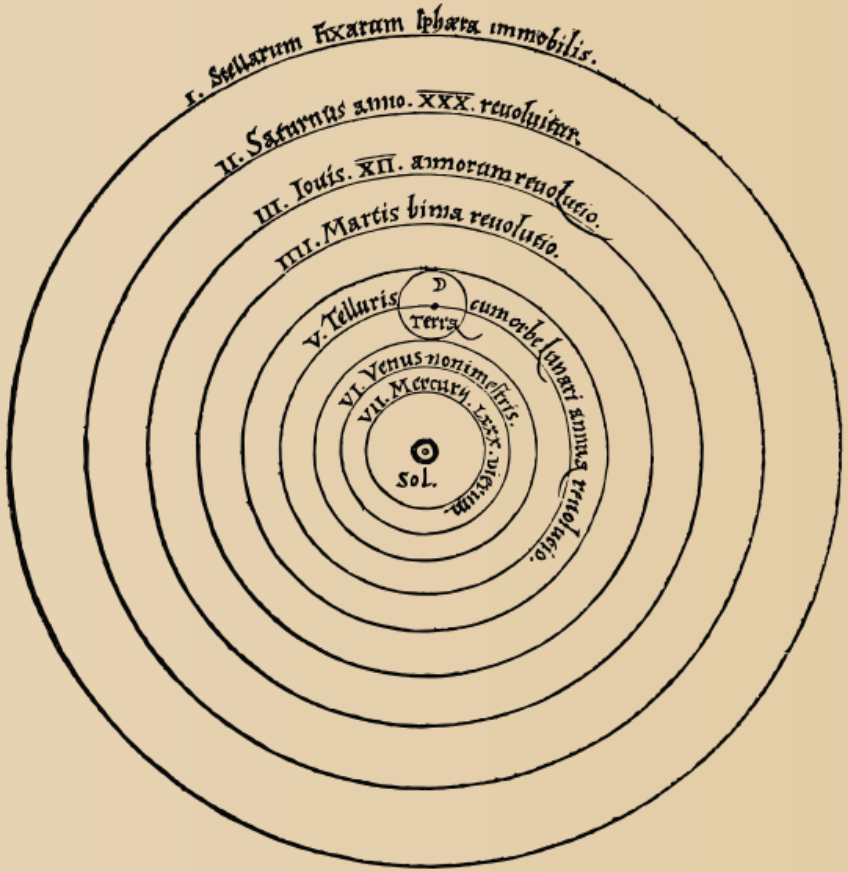
Copernicus was a very unlikely revolutionary; however, the result of his insistence of scientific measurement as a source of truth may well have been the starting point of the coming Scientific Revolution.

https://www.biography.com/video/nicolaus-copernicus-beyond-the-big-bang-18188867655

Copernican Solar System

One of the greatest problems that Copernicus faced was that to get the math to work.

According to his model the Earth had to have two movements, one around the sun and the other around its axis.



Copernican Model

1. There is no one center of all the celestial circles or spheres.
2. The center of the earth is not the center of the universe, but only of gravity and of the lunar sphere.
3. All the spheres revolve about the sun as their midpoint, and therefore the sun is the center of the universe.
4. Closer objects circle faster around the sun than the farther ones.

Copernican Model

5. Whatever motion appears in the firmament arises not from any motion of the firmament, but from the earth's motion. The earth together with its circumjacent elements performs a complete rotation on its fixed poles in a daily motion, while the firmament and highest heaven abide unchanged.

6. What appear to us as motions of the sun arise not from its motion but from the motion of the earth and our sphere, with which we revolve about the sun like any other planet. The earth has, then, more than one motion.

7. The apparent retrograde and direct motion of the planets arises not from their motion but from the earth's. The motion of the earth alone, therefore, suffices to explain so many apparent inequalities in the heavens.

The Gregorian Calendar

The Julian calendar (established by Julius Caesar) divided the year into exactly 365.25 days. This is slightly too long and caused Easter and other Holy Days to slowly drift forward.

Holy days were very important to the Church, and they needed to be computed correctly. Copernicus was invited to Rome to try to produce a more accurate calendar.

The Julian calendar has three consecutive years of 365 days each followed by a leap year of 366 days making the year average 365.25 days.

The Gregorian calendar (365.2425 days) was approved by Pope Gregory in 1582 based upon Copernicus's calculations.

Giordano Bruno 1548–1600

Bruno, an astronomer and mathematician, believed and taught that Copernicus was right. According to Bruno the sun was just a star and there are millions of planets with intelligent beings.

The Church ordered him to stop teaching these ideas and when he didn’t the Inquisition lured him to Venice and got killed.

According to Bruno: “Truth does not change, because it is believed, or not believed, by the majority of the people.”

Conflict

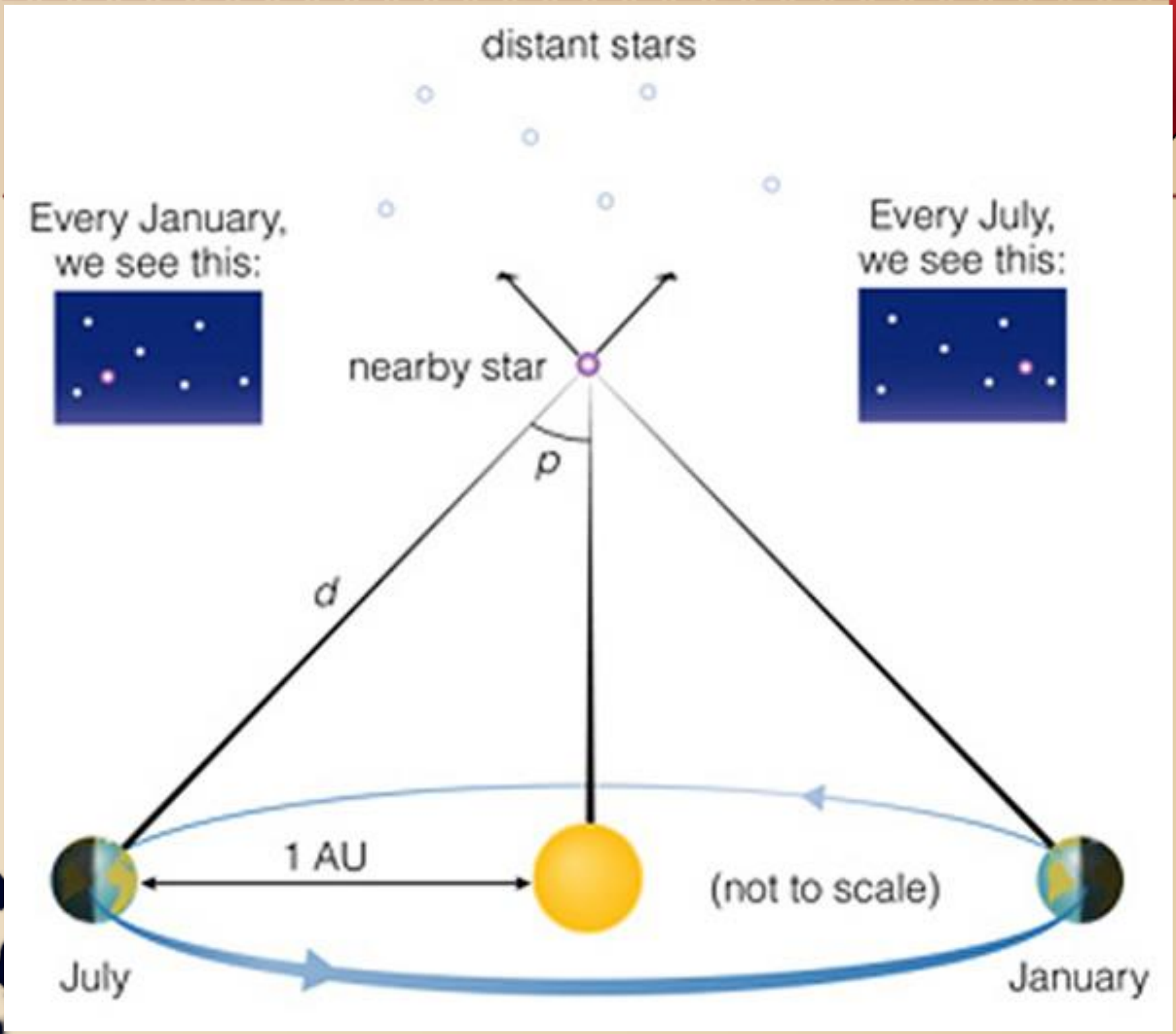
Conflict with the Church over Copernican astronomy.

- Chronicles
  - The world is firmly established; it cannot be moved.
- Psalm:
  - The world is established, firm and secure.
  - God set the earth on its foundations; It can never be moved.
- Ecclesiastes:
  - The sun rises and the sun sets, and hurries back to where it rises.

Tycho Brahe (1546–1601)

Tycho’s observations over thirty years were so accurate that modern instruments and telescopes have largely confirmed his measurements.

To determine the velocity of a body he used a fix star as an independent reference point. However, 6 months of observation did not yield any detectable angular difference, hence he concluded that the earth is stationary.


distant stars
Every January,
we see this:
Every July,
we see this:
nearby star
p
d
1 AU
(not to scale)
July
January

Tycho Brahe (1546–1601)

Late in his life, Tycho realized that he would not be able to analyze the vast amount of data he had collected.

He recruited Johannes Kepler (1571–1630), a German mathematician, to assist him in the calculations.

Johannes Kepler (1571–1630)

Because of the spin of the Earth, a universe of perfect circles made sense from a geocentric perspective.

But in a geocentric world, there is a big problem with Mars. Kepler was able to make the orbit of Mars circular within 8 minutes of arc but Tycho's observations were accurate within 4 minutes of arc.

Finally, he said to himself, Let's go with the Sun. And he gave up all attempts to find a circular motion independent from the Sun as a center.

Elliptic Motion

Kepler (like Archimedes) had divided the circle into infinitely many triangles to find the ratio between the circumference and the diameter of the circle and π.

Presuming that Mars orbited the Sun, Kepler divided the observed orbit of Mars into triangles. Then he calculated the time it took Mars to pass through given segments of the arc.

What he discovered was that Tycho's data were not consistent with a circular orbit, but they were consistent with an elliptical orbit.

Cosmic Whirlpool

An elliptical movement suggested that some kind of force acted on the planets; a force that was weaker with greater distance from the Sun. Kepler's solution was ingenious. He postulated that the Sun rotated creating a vortex, a cosmic whirlpool, that carried the planets with it.

Then he learned about magnetism and theories of the Earth's magnetism. Kepler reasoned that if the Earth and the other planets were large magnets, then the Sun was probably a large magnet, and thus what held the universe together was the simultaneous attraction and repelling of the various magnets on each other.

Orbital Time

Starting in 1609 Kepler published his laws of motion that he had extracted from Tycho's data.

1st Law - The Law of Ellipses - Planets are orbiting the sun in a path described as an ellipse.

https://www.youtube.com/watch?v=qDHnWptz5Jo

Orbital Time

2nd Law - The Law of Equal Areas - The speed at which any planet moves through space is constantly changing. A planet moves fastest when it is closest to the sun and slowest when it is furthest from the sun. Yet, if an imaginary line were drawn from the center of the planet to the center of the sun, that line would sweep out the same area in equal periods of time.

https://www.youtube.com/watch?v=qd3dIGJqRDU

3rd Law of Harmonies

The Law of Harmonies - provides an accurate description of the period and distance for a planet's orbits about the sun. Additionally, the same law that describes the T^2/R^3 (~ 1) ratio for the planets' orbits about the sun also accurately describes the T^2/R^3 ratio for any satellite.

R = 1AU for distance between Earth-Sun

T = 1 Year

https://www.youtube.com/watch?v=KbXVpdlmYZo

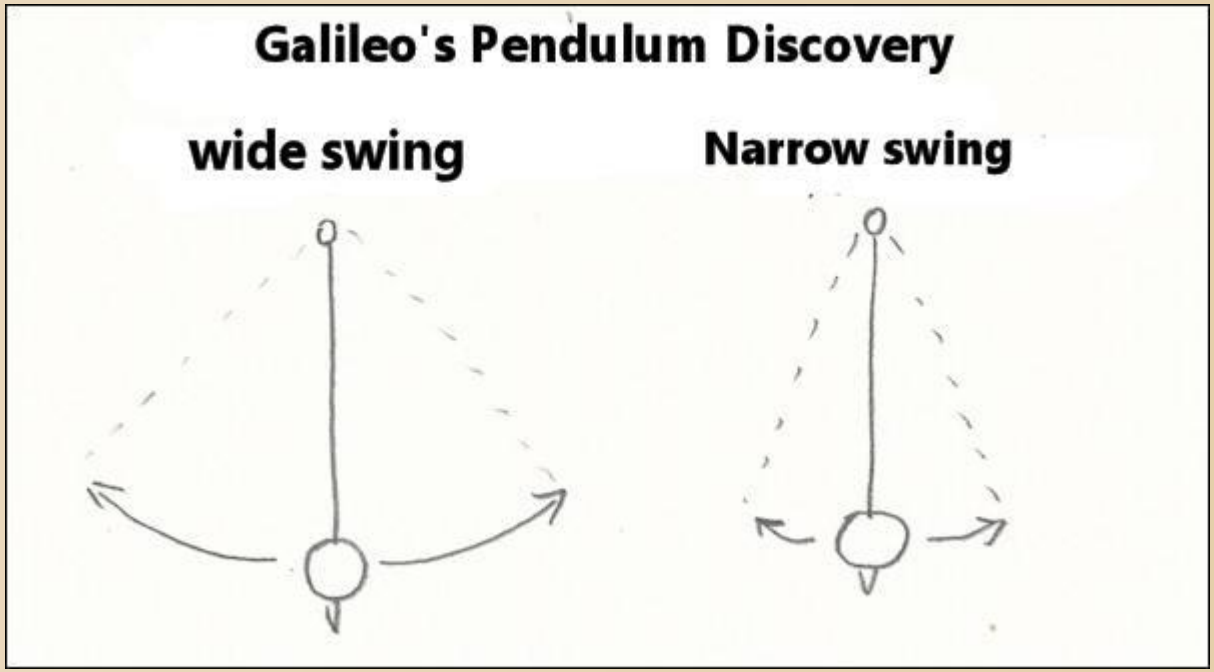
Galileo Galilee (1564–1642 CE)

Galileo’s inquisitive mind would lead him in directions that clashed with the prevailing culture.

He conflicted Aristotle theories and put forward alternative approaches.

On Motion

In 1583 he observed the motions of pendulums, noting that the time of one swing was independent of the height from which the pendulum started the swing.



https://www.youtube.com/watch?v=v2sQF0UFVVE#action=share

Pendulum Clock - 1656

Invented by Dutch scientist Christiaan Huygens,was inspired by investigations of pendulums by Galileo. Galileo discovered that the period of swing of a pendulum is approximately the same for different sized swings.

The pendulum swings with a period that varies with the square root of its effective length.

The introduction of the pendulum, the first harmonic oscillator used in timekeeping, increased the accuracy of clocks enormously, from about 15 minutes per day to 15 seconds per day.

$$T = 2\pi\sqrt{\frac{L}{g}}$$

On Motion

By 1586, Galileo started formulating the laws of falling bodies. He built inclined planes and rolled balls down them to observe their acceleration.

Galileo determined that motion is **Independent of Weight**

It accelerates at constant rate ($g = 9.81$ m / sec^2)

On Motion

The speed of the rolling balls was proportional to the time they fell

- $V = g\,t$

The distance traveled was proportional to the square of the time.

- $\frac{1}{2}\,g\,t^{2}$

He tried different angles of the inclined planes and the mathematical relationships stayed the same. 90 degrees ?

Watch: BBC Documentary – Gravity and Me – Episode I

Galileo's Astronomy

Galileo explained tidal motion in the first clear expression of his belief in the Copernican system of heliocentricity.

Galileo built his own telescope which he made amazing series of discoveries. (Magnify 30 times)

The Moon had craters, mountains, and valleys. Jupiter had four Moons orbiting it in an equatorial plane. The Milky Way was composed of hundreds of unknown stars. Venus had phases just like the Moon. There were dark spots on the Sun.

Two Chief World Systems

He wrote “The Dialogue Concerning the Two Chief World Systems” (Aristotelian and Copernican). The Dialogue initially passed papal censorship, was published (1632), distributed, and quickly became a best seller.

His ideas were not appreciated by the Vatican. The Reformation was well under way and the Church was sensitive to criticism of any kind, real or implied.

Conflict with the Church

Galileo's conflict with the Catholic Church has become a symbol of the conflict between science and religion. Galileo was condemned by the Inquisition in 1633, was forced to renounce his theories, forbidden to publish his books, and placed under house arrest for the remainder of his life.

Galileo believed he was defending the Church against great embarrassment and potential disbelief. Where observation and empirical evidence disputed authority, Galileo believed that authority should give way, even if that authority were Aristotle, or the Church, or the Scriptures themselves.

Revolution of Science - Beginning

While Galileo is not unique, he came to symbolize a new world-view at the beginning of what would be called the Scientific Revolution.

From Galileo's perspective, mathematics not only provided the best understanding of nature, but in many instances, the only understanding of nature.

If one could not understand the language of mathematics, one could not discourse with nature

Transition to Experimental Science

It was a transition that did not happen quickly but was punctuated by important technological developments

- the invention of the telescope and the microscope and accurate devices for measuring temperature, pressure, and time.

In addition, increased knowledge of mathematics also played a role in the development of experimental science.